

CSC 830 - Advanced Artificial Intelligence

Lecture 02: Knowledge Representation and Reasoning - Part I

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Agents that Reason Logically

- Overall agent design
- Basic elements of agent design: a formal language in which knowledge can be expressed, and a means of carrying out reasoning in such a language – these 2 elements constitute logic.

- A knowledge-based agent consists of a knowledge base (KB) and an inference engine (IE).
- A knowledge-base is a set of representations of what one knows (facts) about the world (objects and classes of objects, the fact about objects, relationships among objects, etc.)
- Each individual representation is called a sentence.
- The sentences are expressed in a knowledge representation language.
- Examples of sentences
 - The moon is made of green cheese
 - If A is true then B is true
 - A is false
 - All humans are mortal
 - Confucius is a human

- The Inference engine derives new sentences from the input and KB
- The inference mechanism depends on representation in KB
- The agent operates as follows:
 1. It receives percepts from environment
 2. It computes what action it should perform (by IE and KB)
 3. It performs the chosen action (some actions are simply inserting inferred new facts into KB).

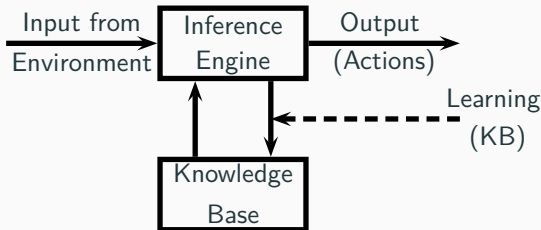


Figure 1

KB can be viewed at three different levels

- **Knowledge Level.**

- The most abstract level – describes agent by saying what it knows.
- Example: A taxi agent might know that the River Niger Bridge connects Onitsha with Asaba.

- **Logical Level.**

- The level at which the knowledge is encoded into sentences.
- Example: `Links(RiverNigerBridge, Onitsha, Asaba)`.

- **Implementation Level.**

- The physical representation of the sentences in the logical level.
- Example: `"(Links RiverNigerBridge, Onitsha, Asaba)"`

Knowledge Representation and Reasoning

Introduction

- The objective of knowledge representation is to express the knowledge about the world in a computer-tractable form, so that agents can perform well. Many Knowledge Base systems rely on some variant of logic.

The objective of knowledge representation is to express knowledge in a computer-tractable form, so that agents can perform well. A knowledge representation language is defined by:

- Its **syntax** which defines all possible sequences of symbols that constitute sentences of the language (grammar to form sentences)
- Its **semantics** determines the facts in the world to which the sentences refer (meaning of sentences) Each sentence makes a claim about the world.
- Its **proof theory** (inference rules and proof procedures)

Knowledge Representation Schemes

In AI, there are four basic categories of representational schemes: logical, procedural, network and structured representation schemes.

- 1) **Logical representation** uses expressions in formal logic to represent its knowledge base. Predicate Calculus is the most widely used representation scheme.
- 2) **Procedural representation** represents knowledge as a set of instructions for solving a problem. These are usually if-then rules we use in rule-based systems.
- 3) **Network representation** captures knowledge as a graph in which the nodes represent objects or concepts in the problem domain and the arcs represent relations or associations between them. Semantic networks and conceptual graph are example of this scheme
- 4) **Structured representation** extends network representation schemes by allowing each node to have complex data structures named slots with attached values, theses values may be simple numeric or symbolic data, pointer to other frames or even procedures, scripts and frames are examples of this scheme. In this lecture we shall

Knowledge Representation and Reasoning via Logic

Logical representation

: A formal language for expressing knowledge and ways of reasoning.

Logic is defined by:

- A set of sentences – A sentence is constructed from a set of primitives according to syntax rules.
- A set of interpretations – An interpretation gives a semantic to primitives. It associates primitives with values.
- The valuation (meaning) function V – Assigns a value (typically the truth value) to a given sentence under some interpretation V :
 $\text{sentence} \times \text{interpretation} \rightarrow \{\text{True}, \text{False}\}$

For 2000 years, people tried to codify “human reasoning” and came up with logic.

- Most AI work until 1980s: Build machines that represent knowledge and do reasoning via logic. “Rule based reasoning.”
- “Learning from data” is popular today, but lacks aspects that were trivial in the pre-1980s systems (e.g. allow human programmer to easily communicate his/her knowledge to the system). “How do you teach a deep net to multiply two numbers?”
- Logical reasoning now seems poised for a comeback.

Goals of logic

- Represent knowledge about the world.
- Reason with that knowledge.

Natural Language

- 500 Naira is better than 200 Naira.
- 200 Naira is better than 100 Naira
- Therefore, 500 Naira is better than 100 Naira
- 100 Naira is better than nothing.
- Nothing is better than world peace.
- Therefore, 100 Naira is better than world peace.

Natural language is tricky!

Use of logic removes ambiguity (similar to computer languages); but also makes system less flexible. (Will study more flexible versions later.)

Components of any logical system

- Syntax
- Semantics.
- Reasoning (Inferencing)

Propositional Logic (aka Boolean Logic)

Syntax:

- **Propositional symbols (atomic formulas):** $A, B, C, \dots$
- **Logical connectives:** $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
- **Build up formulas recursively** - if f and g are formulas, so are the following:
 - **Negation:** $\neg f$
 - **Conjunction:** $f \wedge g$
 - **Disjunction:** $f \vee g$
 - **Implication:** $f \rightarrow g$
 - **Biconditional:** $f \leftrightarrow g$

$\vee$ and $\wedge$ and $\leftrightarrow$ are symmetric, like “+” and “times” in arithmetic.

- Syntax provides symbols.
 - No “meaning” yet (semantics)!
- Semantics provided by a “Model” (unrelated to “model” used in machine learning!)
- For propositional logic, a model is simply an assignment to all variables. (each variable assigned 0 (false) or 1 (true), not both)
- Check: What is # of possible models if there are 3 variables? How about n variables?

- Need an **interpretation** of symbols for a given set of sentences
 - Proposition symbols do not have meaning by themselves
 - An interpretation connects proposition symbols to a statement about the world (which may be true or false in that world)
 - An interpretation in PL can be defined as an **assignment of truth values** to all proposition symbols involved
 - There are many interpretations for a given set of sentences (2^n if they involve n distinct proposition symbols)

Interpretation Function

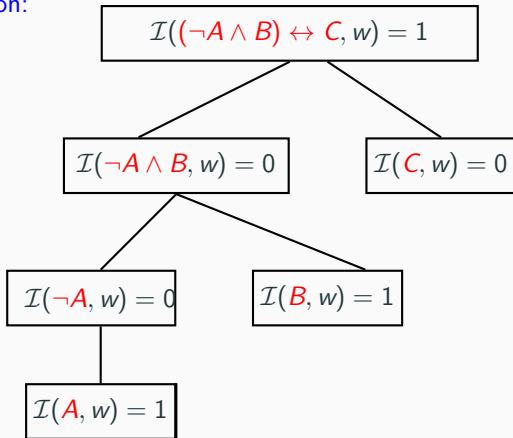
- $I(f, w)$: Given formula f and model w , assigns exactly one of 1 (True) or 0 (False) to f .
- Build up formulas recursively - if f and g are formulas, so are the following:
 - Negation: $\neg f$, true iff f is false
 - Conjunction: $f \wedge g$, true iff both f and g are true
 - Disjunction: $f \vee g$
 - Implication: $f \rightarrow g$
 - Biconditional: $f \leftrightarrow g$

Example

Formula: $f = (\neg A \wedge B) \leftrightarrow C$

Model: $w = \{A : 1, B : 1, C : 0\}$

Interpretation:



Definition

:

$$M(f) = \text{Set of models } w \text{ for which } I(f, w) = \text{True}$$

Formula f compactly represents $M(f)$ ("Set of possible worlds where f is true.")

Example: For $f = A \wedge B$.

$$M(f) = \{A = 1, B = 1\}.$$

For $f = A \leftrightarrow B$

$$M(f) = \{A = 1, B = 1\}; \{A = 0, B = 0\}$$

- **Tautology:** Formula f such that $M(f) = \text{All possible models}$. ("True in all possible worlds")

Example: $A \vee \neg A$ (True whether $A = 0$ or $A = 1$!)

- **Contradiction:** Formula f such that $M(f) = \text{Empty set}$. ("False in all possible worlds.")

Example: $A \wedge \neg A$ (False whether $A = 0$ or $A = 1$)

Knowledge Representation via logic

- Knowledge base : Set of formulae $\{f_1, f_2, \dots, f_n\}$

$M(KB) =$ All possible models for $f_1 \wedge f_2 \wedge \dots \wedge f_n$.

- Formulae = “known facts ”
- Models = all possible “worlds” where all these facts hold (Adding more facts to KB can only shrink set of possible worlds.)

Example: R, S, C (“Rainy”, “Sunny”, “Cloudy”)

KB $R \vee S \vee C$; (“It is either Rainy or Sunny or Cloudy.”)

$R \rightarrow C \wedge \neg S$; (“If it is Rainy then it is Cloudy and not Sunny.”)

$C \leftrightarrow \neg S$ (“If it is Cloudy then it is not Sunny, and vice versa”)

Models for KB: $\{R=1, S=0, C=1\}; \{R=0, C=1, S=0\}; \{R=0, C=0, S=1\}$.

- Defn: Knowledge-base KB is satisfiable if $M(KB) \neq \emptyset$ (i.e. there is some assignment to variables that makes all formulae in KB evaluate to True)
- Defn: KB contradicts formula f if $KB \cup \{f\}$ is not satisfiable
- Defn: KB entails formula f (denoted $KB \models f$) if $M(KB \cup \{f\}) = M(KB)$. (in every world where KB is true, f is also true)
- Defn: KB is consistent with formula f if $M(KB \cup \{f\})$ is non-empty (there is a world in which KB is true and f is also true)

Check: KB entails f iff it contradicts $\neg f$.

An Example

Example: Variables: R, S, C (“Rainy”, “Sunny,” “Cloudy”)

$$\begin{aligned} KB: \quad & R \vee S \vee C; \\ & R \rightarrow C \wedge \neg S; \\ & C \leftrightarrow \neg S \end{aligned}$$

Does $KB \models S \wedge C$?

Models for KB : $\{R = 1, S = 0, C = 1\}; \{R = 0, C = 1, S = 0\}; \{R = 0, C = 0, S = 1\}$.

$S \vee C$ is true in all these models.

An Example

Example: Variables: R, S, C (“Rainy”, “Sunny,” “Cloudy”)

$$\begin{aligned} KB: \quad & R \vee S \vee C; \\ & R \rightarrow C \wedge \neg S; \\ & C \leftrightarrow \neg S \end{aligned}$$

Add a variable: U (“Carry an umbrella”). What common-sense “facts” can we add about U to the above KB?

Examples: $R \rightarrow U; S \rightarrow \neg U$

- AI systems till 1980s used such reasoning; “facts” were added by programmers based upon introspection.
- Decision-making at run-time = which formulae are entailed/contradicted/consistent

Truth table method (to check if KB has any model)

- If n variables, can take 2^n time. (infeasible for even $n = 100$) Any faster algorithm?
- Polynomial time algorithm ($P = NP$ (Famous open problem))
- In practice there are reasonable algorithms that use resolution and other related reasoning methods.

- Logical Inference is used to create new sentences X that logically follow from a given set of sentences S (in KB and from input), i.e., $S \models X$
- This kind of inference is also known as deduction
- Use truth table: Whether $S \models X$ holds can be determined by whether $S \Rightarrow X$ is a tautology
 - $S \models X$ holds iff every interpretation I that makes S true also makes X true
 - $S \Rightarrow X$ is a tautology iff every interpretation I that makes S true also makes X true
 - Example: $P, P \Rightarrow Q \models Q$ because $(P, P \Rightarrow Q) \Rightarrow Q$ is true under any interpretation
 - Huge truth tables for inference involving large number of sentences

Resolution procedure to decide satisfiability of a KB

(simplest version; [Davis-Putnam, 1950s])

- KB consists only of formulae that are **clauses** (ie $\vee$ of variables or negated variables). (With some work, can convert every KB to this form.)
- Warmup: What can we conclude under foll. conditions?
 - KB has singleton clauses (**A**), ($\neg$ **A**).
 - HAS NO MODEL (UNSATISFIABLE)!
 - KB contains clause pairs of form (**A** $\vee$ $B_1 \vee \dots \vee B_n$) and ($\neg$ **A** $\vee$ $C_1 \vee \dots \vee C_m$)
 - Every model for KB must make $(B_1 \vee \dots \vee B_n \vee C_1 \vee \dots \vee C_m)$ TRUE

While KB nonempty **do**

{

If KB contains clause pairs of form $(A), (\neg A)$

Print ("No Model.") and **STOP**. /*WHY?*/

If KB contains clause pairs of form $(A \vee B_1 \vee \dots \vee B_n)$ and $(\neg A \vee C_1 \vee \dots \vee C_m)$

 Add $(B_1 \vee \dots \vee B_n \vee C_1 \vee \dots \vee C_m)$ to KB. /*WHY?*/

else

Print ("Model exists") and **STOP**. /* WHY??*/

}